

BrainBook Final Report

A Stimulating, Personalized Guide for the Modern Age

BrainBook

Keywords: Write here... Write here..

Question: Write a new Question here...

Prior Knowledge: Describe what you
already know about the topic here...

Open

Existing

Gautam Khajuria, Sarvesh Jayaraman

CMPCS 185, Winter 2026

—

thebrainbook.github.io

1. Project name & value proposition

BrainBook is a stimulating, personalized guide for the modern age. Instead of acting like a shortcut to answers, BrainBook helps students work through difficult material more actively. It is designed to support thinking, not replace it.

2. Team Members

Sarvesh Jayaraman - Designer, Actor, & Researcher

Gautam Khajuria - Designer, Editor, & Researcher

3. Problem/solution overview

Current AI tutoring often collapses learning into “ask to get an answer,” which reduces student agency and skips the step where students externalize and debug their own reasoning. Many users also felt current AI tools were too direct, too agreeable, and too quick to take over the thinking process. BrainBook solves this with a neutral, adaptive AI textbook that gives small, digestible hints and then asks targeted follow-up questions so students build the explanation themselves. This makes learning feel more active and earned.

4. Needfinding

Interviews

To understand AI as a learning tool, we interviewed users in academic and professional settings. Our initial needfinding focused on AI use in education and the workplace, major contexts where AI supports learning. We aimed to understand how students use AI for general concepts and how professionals use it for specialized job knowledge. Key questions guided by these interviews were what participants use AI for most, what they want from it as a learning tool, what features would increase ownership, and what prompting habits or tools they use.

In this first round, we interviewed Fanbo, an 18-year-old English major at SBCC, Jane, a 20-year-old biology student in the UK, and Sam. Fanbo described AI as more useful for finding sources than for actually teaching, and he did not feel AI had made him a better learner. He also emphasized that he checks sources carefully, dislikes agreeable AI behavior, and believes the best learning tool should challenge him rather than flatter him. Jane shared similar frustrations, saying that AI often gives answers too directly, feels too polite, and does not support real retention. She wanted something more natural that would ask what the learner had already tried

instead of simply dumping the solution. Sam added a different-yet-related concern: trust. He saw AI as useful in smaller support roles, but felt accuracy and verification mattered enough that he would even accept slower responses if they were more reliable. Across these early interviews, we saw the same pattern repeatedly: users did not want AI to replace their thinking, and they did not fully trust it when it acted too confidently or too passively.

In GR2, we further distinguished between AI that teaches and AI that simply does work by interviewing two users. Nikola, a 37-year-old software engineer and father, viewed AI as a flexible, conversational work tool and preferred a system that both assists and teaches. Ellie, a 20-year-old psychology student, used AI as a research booster, not a replacement for thinking, and wanted a coach-like system that asks better follow-up questions, though she noted that confident AI responses could cause self-doubt. These interviews guided us toward a design opportunity, building an AI system that supports learning through guidance, challenge, and reflection, rather than by simply providing immediate answers.

Synthesis

After completing our needfinding interviews, we stepped back to identify common themes across participants with different academic and professional backgrounds. We created empathy maps for each interviewee rather than forcing all of their experiences into a single synthesis, which helped us better understand not just what users said about AI, but how they actually approached learning, confusion, and trust in practice. During this phase, we found several recurring themes that later shaped our HMWs and POVs, including the need for AI to challenge users more, support retention instead of shortcutting the process, and help learners feel more confident in their own reasoning. Throughout synthesis, we kept asking how each insight connected across interviews, and by repeatedly revisiting and refining these patterns, we were able to identify the deeper needs behind BrainBook.

5. POVs & experience prototypes

We wrote POVs for interviewees Ellie and Nikola. Our selected POVs focused on two needs we saw most clearly: helping users stay engaged in the learning process, and helping them trust their own reasoning while using AI.

POV #1

One selected POV came from Ellie's experience using AI to simplify confusing class content and save time while studying. We were surprised to realize that when AI makes learning too easy, it can reduce motivation and make studying feel finished rather than rewarding. We wondered if the problem was not just accuracy, but that AI often removes the struggle that makes learning stick. It would be game-changing to make AI feel like a coach that builds understanding and curiosity without doing the work for the student.

The POV:

1. We met a psychology student who uses AI to clarify research papers and identify broader implications for research direction.
2. We were surprised to learn that confident AI answers can conflict with what she already knows and make her doubt her own judgment.
3. We wonder if this means that trust and confidence, rather than access to information, are the main bottlenecks in using AI for research help.
4. It would be game changing to help students use AI for research synthesis while keeping their confidence and judgment intact.

POV #2

A second selected POV also came from Ellie. We met a psychology student who uses AI to clarify research papers and identify broader implications for research direction. We were surprised to learn that confident AI answers can conflict with what she already knows and make her doubt her own judgment. We wondered if trust and confidence, rather than access to information, were the main bottlenecks in using AI for research help. It would be game-changing to help students use AI for research synthesis while keeping their confidence and judgment intact.

The POV:

1. We met a student who uses AI to simplify confusing class content and save time when studying.
2. We were surprised to realize that when AI makes learning too easy, it can reduce motivation and make studying feel finished rather than rewarding.
3. We wonder if this means that the problem is not just accuracy, but that AI often removes the struggle that makes learning stick.
4. It would be game changing to make AI feel like a coach that builds understanding and curiosity without doing the work for the student.

POV #3

Our third selected POV came from Nikola, a software engineer and homeschooling father. We were surprised to learn that he saw AI as having many of the same shortcomings that people do. We wondered if many of the problems AI has while teaching are really interaction issues that a better design could reduce. It would be game-changing to mitigate those errors and make AI-supported learning feel more natural and adaptive.

The POV:

1. We met Nikola, a 30-year-old software engineer at Box, Inc. and a homeschooling father of 2

2. We were surprised to learn that Nikola notes that AI has the same shortcomings that people do
3. We wonder if this means that many of the problems that AI has while teaching are more so user errors
4. It would be game-changing to try to mitigate these user errors

Sampling of the HMWs that stemmed from each POV

From these POVs, we developed HMWs that focused on curiosity, personalization, and confidence. One asked how we might help users stay curious and still feel the reward of effort after getting an explanation. Another asked how we might adapt to different students' learning styles, needs, and responsibilities while staying personal. A third asked how we might help users build a habit of using their own judgment and intuition before accepting an explanation so that they trust themselves more. These HMWs helped us move from broad insights about AI tutoring to specific design opportunities.

Our Top Three Solutions

1. An open-world game that explains topics by treating each one as a quest that sends users across the map to unlock varying explanations and quizzes
2. An argumentative AI that "fights" the user to gain further context, stimulate the user, and target the specific areas that they need help with using adaptive teaching modes
3. A contradiction detector that asks what you thought before and then flags where the AI's answer disagrees with that belief.

Brief description of each experience prototype

1. Open-world learning game

i. The assumption being tested

This prototype tested whether a game-like learning structure would keep students engaged through the full learning process and make explanations feel more rewarding.

ii. Key aspects of the prototype setup

We created a Google Doc that framed the learning experience as a quest, with time constraints, goals, and a sequence of text-based RPG-style prompts and questions. The prototype was designed to make learning feel exploratory rather than static.

iii. What worked/didn't work, implications

Users found the format engaging and liked the freedom it gave them. However, the open structure did not always push them to reflect on what they already knew, and one user described it as easy to skim. This suggested that curiosity alone was not enough, and that the final experience would need more structure and clearer checkpoints for reasoning

2. Argumentative AI

i. The assumption being tested

This prototype tested whether an argumentative AI that asks questions first would be able to fully gauge the user and determine what adaptive teaching modes to use.

ii. Key aspects of the prototype setup

We used a Google Doc simulating an AI chatbot entry window. The user typed in a query, and the simulated AI, played by Gautam, responded. The first response asked the user to explain their existing understanding. The AI did not immediately give a full response, and instead gave little bits of information before asking the user to think more.

iii. What worked/didn't work, implications

This AI strategy worked well, and the user was very satisfied and learned strongly. The constant prompting forced them to continue thinking about what they already knew, and one user described it as “almost a guided explanation to myself.” It also helped fill in gaps in their knowledge. Since our assumption was verified, we decided this approach would likely be integrated into our final solution.

3. Contradiction detector

i. The assumption being tested

This prototype tested whether asking users for their own answer first, then comparing it with the AI's answer, would help them reflect more deeply and trust their own reasoning.

ii. Key aspects of the prototype setup

We again used a Google Doc simulating an AI chatbot. Users first explained their own understanding, then the AI generated its response, highlighted differences between the two answers, and explained where the disagreement came from.

iii. What worked/didn't work, implications

Users felt this prototype helped them learn from their mistakes, and the visual highlighting made gaps in their reasoning easier to see. This suggested that comparison and reflection should be an important part of the final design, especially for helping users learn without losing confidence.

6. Design Evolution

Final solution

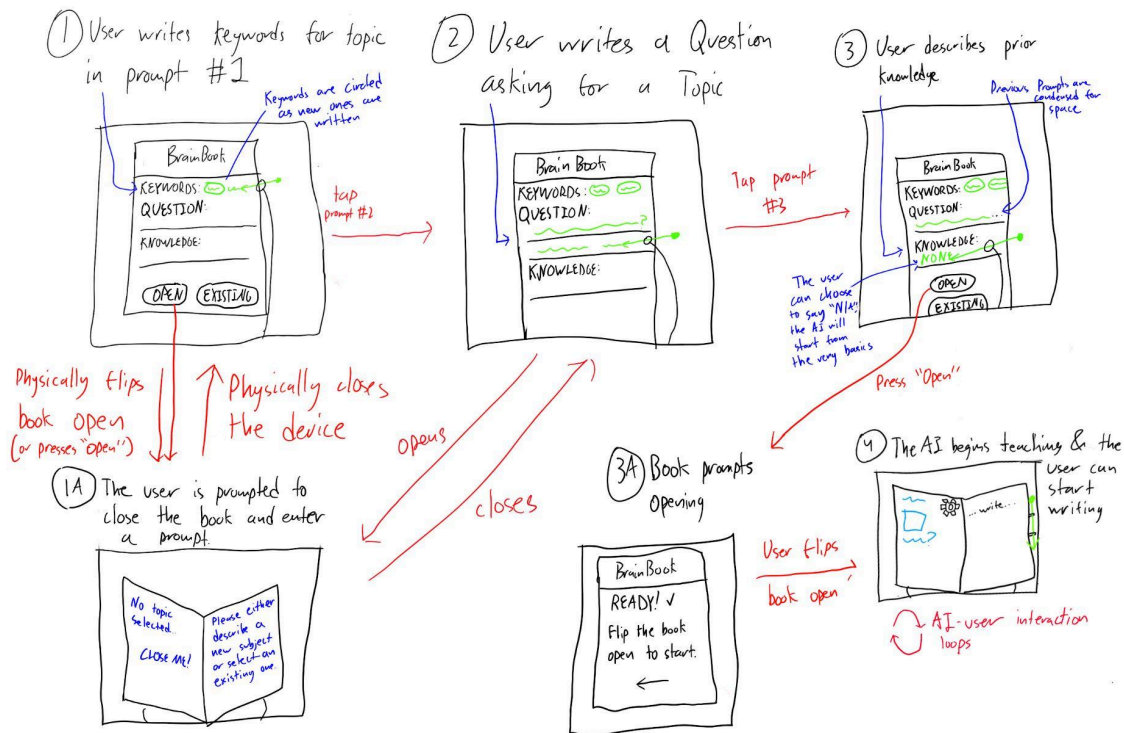
i. Description

BrainBook's final solution is a two-screen, textbook-like AI learning interface that helps students work through a question by separating AI Feedback from My Responses. The system begins with a structured cover page where users enter keywords, their question, and prior knowledge, then moves into a guided learning session where the AI gives small hints and the student builds their own explanation step by step. In the high-fidelity version, the product also includes onboarding, settings for preferred learning style and favorite learning tools, numbered links between AI feedback and user responses, session-completion feedback, and navigation controls for moving across reset pages in longer sessions.

ii. Rationale for the selected solution, grounded in evidence

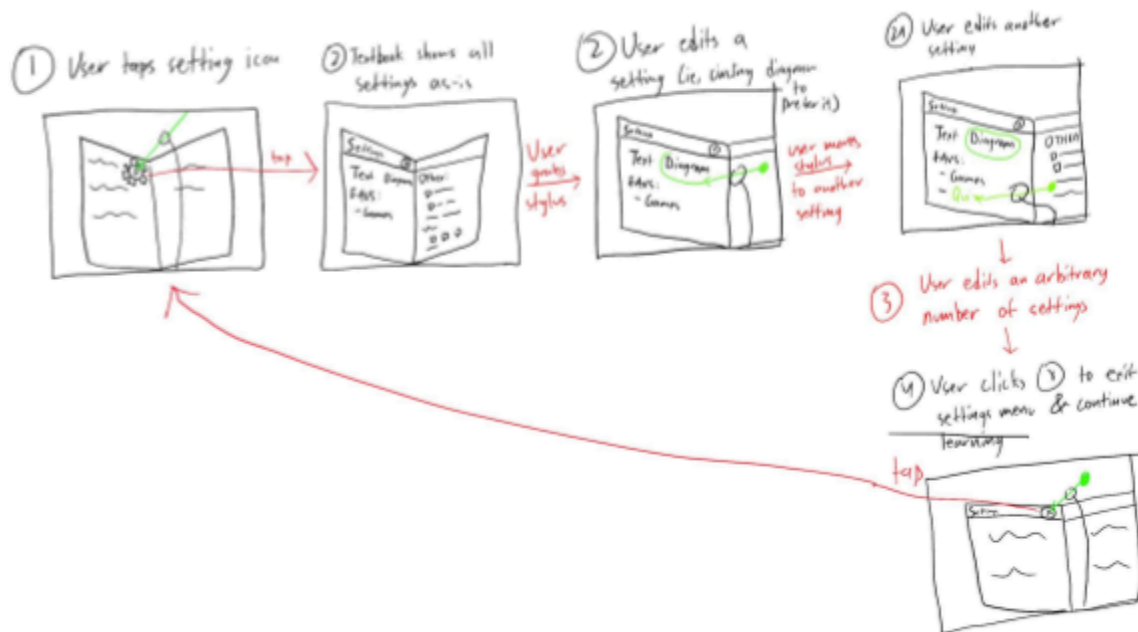
We selected this solution because it best carried forward the core direction of BrainBook while also addressing the major usability issues found in later testing. Earlier needfinding and experience prototypes showed that users wanted AI to act more like a coach than an answer machine, so the final design kept the step-by-step hinting structure and the emphasis on user-generated reasoning. The textbook-like design further emphasized this. At the same time, our high-fidelity evaluations showed that the concept needed much stronger structure and clarity to work well as an interface.

Tasks



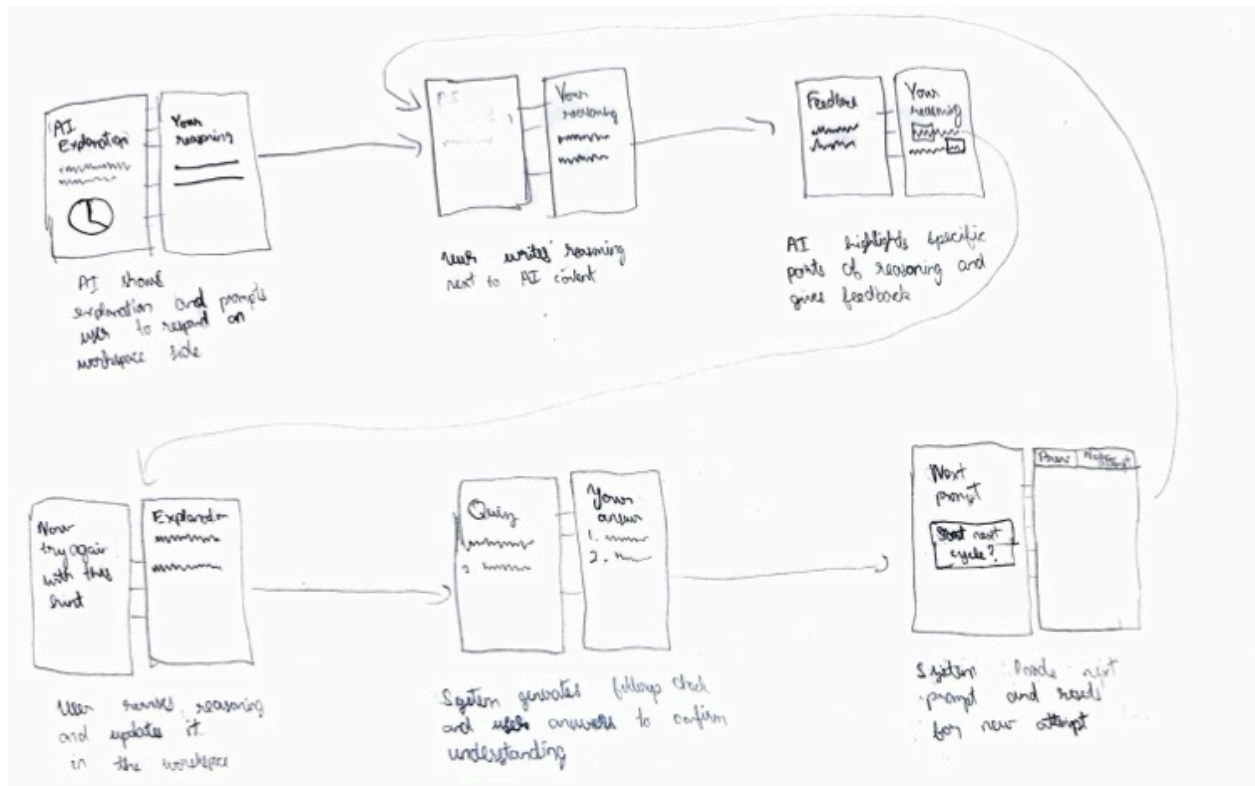
Simple task: open the textbook and begin a session

The user begins on BrainBook's opening flow by entering a few starting inputs, such as keywords, their question, and prior knowledge, and then opening the textbook to start the session. In the medium-fidelity prototype, this task established the basic entry into the product, while in the high-fidelity prototype, it became clearer through a more structured start screen and onboarding that explained how the interface works. This task is important because every user must complete it before any learning can take place, and it allows BrainBook to gather enough context to make the session feel more tailored from the start. In the annotated task flow above, we show the opening screen, the areas where the user provides initial information, and the transition into the main learning interface.



Moderate task: personalize the experience through settings and learning preferences

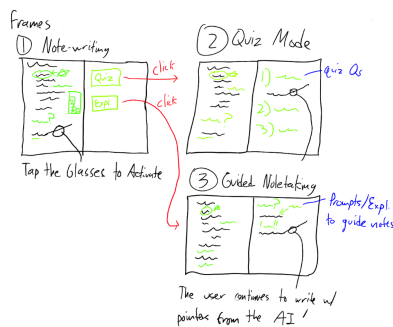
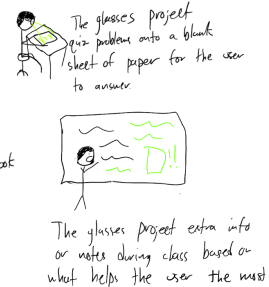
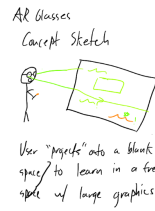
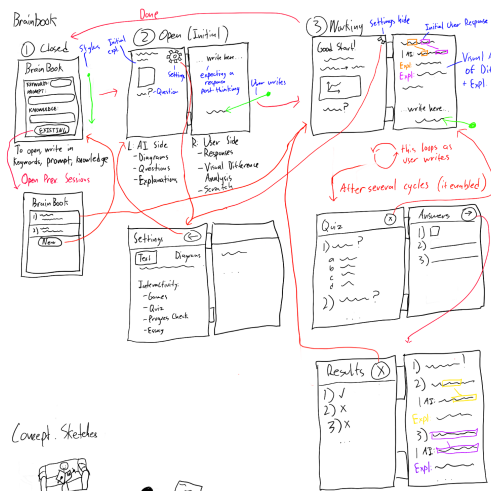
This task allows users to shape BrainBook around how they learn best. In the medium-fidelity prototype, users could open the settings page, select preferences such as diagrams and favorite learning tools, and return to the main session. In the high-fidelity prototype, this flow became more understandable through clearer labels and a save-state dialog that made it more obvious when settings changes were being kept. This task is important because, although BrainBook can still function without personalization, repeat users benefit from an experience that better matches their study habits and preferred learning supports. In the annotated task flow above, we show the settings screen, the adjustable preferences, and the return to the main learning session.



Complex task: move through the main learning loop by reading AI feedback and writing responses

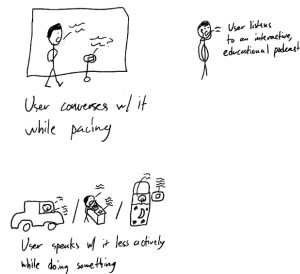
This task is the core BrainBook experience. After opening the textbook, the user reads short AI feedback on one side and writes their own response on the other, continuing through the session step by step. In the medium-fidelity prototype, this interaction was represented through the two-screen structure, and in the high-fidelity prototype, it was refined with clearer placeholders, numbered links between AI feedback and student responses, and controls for navigating across reset pages. This task is especially important because it is where the actual learning happens. Instead of receiving one large answer, the user builds understanding through smaller hints, reflection, and follow-up reasoning. In the annotated task flow above, we show how the user moves between AI feedback and their own response space, and how the interface supports this repeated cycle of guided explanation-building.

Design Evolution



Voice Menu

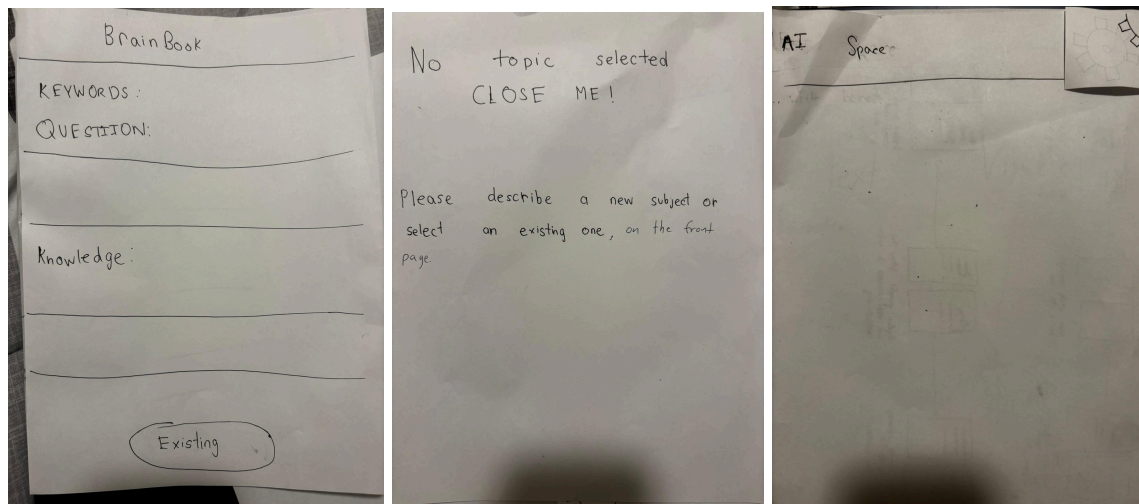
Concept Sketches



Our Three Original Concept Sketches & Key Frames

Sketches

Before deciding on the modality of our solution, the team sketched several possible directions for BrainBook. Ultimately, we settled on a textbook-style two-screen interface because it felt like the most straightforward way to support guided learning. Additionally, it clearly distinguished between the user and AI's responses, which was a priority of our design to ensure user agency.

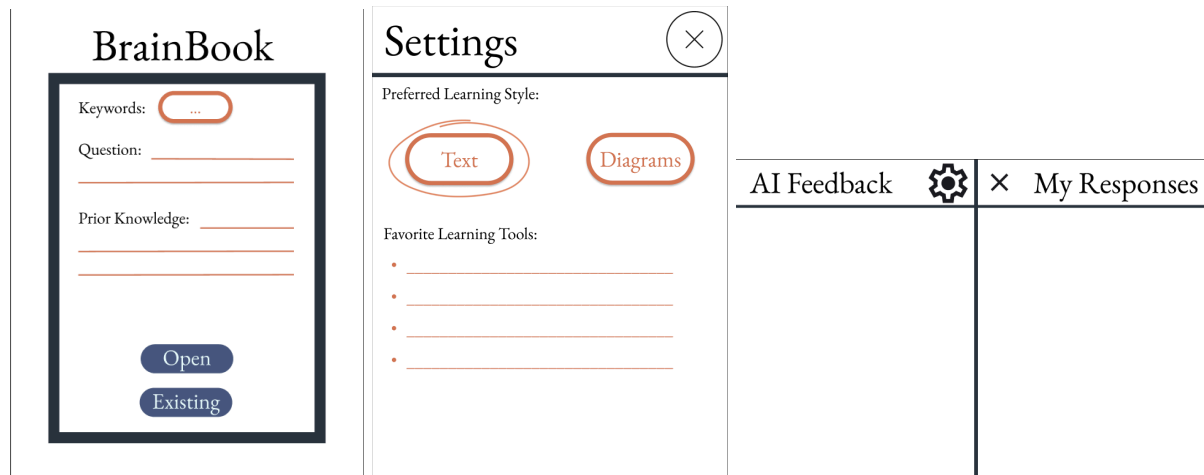


Some Screens from our Low-fi Prototype

Low-fi prototype

After choosing the device direction, we built a paper prototype to evaluate the core interaction structure.

In-person interviews were conducted using this prototype, with both the user and the "AI" writing on the physical pages/screens. This low-fidelity iteration tested the product's fundamental concept: whether users could understand the textbook metaphor, start a session, and follow a guided explanation easily. The paper prototype allowed high-level concept assessment before refining the UI.



Some Screens from our Med-fi Prototype. We mostly emulated the Low-fi prototype, but added extra elements (like the 'Open' button) for extra clarity on how to navigate the prototype.

Medium-fi prototype

We moved from a basic concept to a medium-fidelity Figma prototype to test the near-final interface. This clarified team tasks, usability goals, and necessary revisions. In GR6, we refined three tasks (opening the textbook, updating it with user reasoning, and personalizing AI responses) and two usability goals (intuitive textbook interaction and reduced reasoning errors). We evaluated our low-fi prototype through user testing, which showed a lack of clarity on how to start the application, use the Settings page, and stop the application. We directly addressed these issues by focusing on a clearer start, improved session control, distinguished user/AI input, and simplified settings. Specific changes included a redesigned front page with a visible Open button and clearer writing area, an 'X' button for ending sessions, and consolidated settings.

High-fi prototype & Heuristic evaluation violations

In our studio, we exchanged our medium-fidelity prototypes with other teams to perform heuristic evaluations. These evaluations used Nielsen's 10 Usability Heuristics along with the additional heuristics used in class. BrainBook's medium-fidelity prototype received 78 total violations across three evaluations, with 11 rated as level 4 and 28 rated as level 3 in severity. The most common issues involved visibility of system status, consistency and standards, and recognition rather than recall.

We addressed the most severe prototype violations by reviewing how they failed certain heuristics and deciding whether to fix or intentionally leave the issues.

To resolve weak system feedback, we implemented a session completion pop-up, a warning before page reset, and back/next controls. These additions helped users better understand state changes and revisit previous material.

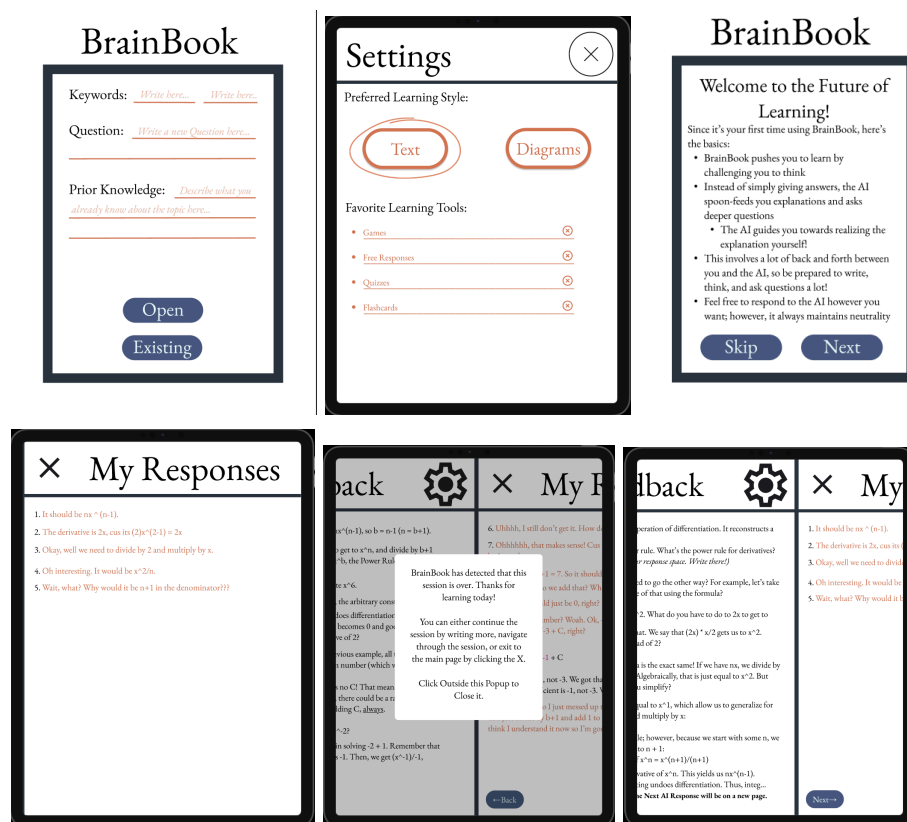
For poor recognition, we added placeholders to the cover page, settings page, and response area, making it easier for users to identify what and where to write.

To clarify the relationship between AI and user content, we numbered related AI feedback and user responses. We also renamed "AI Space" and "User Space" to "AI Feedback" and "My Responses."

Some severe violations remained unaddressed because they stemmed from intentional prototype tradeoffs, not genuine design flaws. For instance, we kept the feature allowing users to tap anywhere to advance the conversation, as this simulated both writing and interaction within the Wizard-of-Oz prototype in Figma.

Similarly, the "Open" button was left active even when fields were blank. This was to model a physical foldable device that can always be opened, even if it leads to an error state. We also chose not to consider the necessity of remembering AI feedback while moving between panels as a flaw, as this reflection was integral to the intended learning process.

These findings directly shaped the final high-fidelity prototype. The visualizations below show the major revisions that resulted from heuristic evaluation, including the redesigned cover page, revised settings page, onboarding guide, numbered feedback-response pairs, session completion popup, and page reset controls. Together, these changes made BrainBook easier to understand, easier to navigate, and more transparent during use.



7. Final prototype

Hard-coded data used

To simulate BrainBook, the prototype hard-coded several features. These included the three initial prompts (Keywords, Question, Prior Knowledge), all AI Feedback responses, and user responses via click-to-fill instead of real text entry. Settings like Favorite Learning Tools and the Existing Interactions page were also prebuilt. This hard-coding allowed us to test the learning experience's structure and usability despite lacking live AI or open-ended writing capabilities.

AI tools used

Although BrainBook is centered around AI-supported learning, the final prototype did not use a live AI model. Instead, AI behavior was simulated through Wizard-of-Oz interactions and prewritten responses. This was appropriate for our stage of development because the goal of the prototype was not to test model quality, but to evaluate the interaction design and learning experience. By simulating the AI in advance, we were able to focus on whether BrainBook's core concept worked, whether students understood the guided flow, and whether the product encouraged active reasoning instead of passive answer collection.

8. Reflection & next steps

Main learnings

One of our biggest takeaways from this quarter was that strong design comes from repeatedly narrowing broad ideas into specific user needs. At the beginning of the project, we were interested in AI and learning in a general sense, but through interviews, synthesis, POVs, and prototyping, we realized that the real issue was not that students needed more answers. They needed a better process for working through confusion without losing ownership of their reasoning. That shift changed BrainBook from a broad AI learning idea into a more focused product centered on guided explanation-building.

We also learned that user feedback often reveals problems that are not obvious when a concept seems strong in theory. Across our prototypes, we saw that even when users liked the core idea, the details of structure, clarity, and flow mattered just as much. The move from medium-fidelity to high-fidelity especially showed us that an educational concept can only succeed if the interface clearly supports the intended interaction. This made us more aware of how much design thinking depends on iteration, not just ideation.

For BrainBook specifically, we learned that a learning tool should not simply make work easier. It should make thinking more visible and more manageable. Our interviews and testing showed that users wanted AI to act more like a coach than an answer machine. That insight stayed consistent throughout the quarter and became the foundation of our final design.

Things to add in the future

If we had more time, the next biggest step would be turning BrainBook from a simulated prototype into a working system with a real AI backend. That would allow us to test how well the guided hinting and follow-up questioning actually adapts to different students, subjects, and levels of prior knowledge. We would also want to explore stronger personalization, such as adjusting the amount of support based on how confident a student is or how much they already know.

Another important next step would be expanding the system's memory and feedback across sessions. Right now, BrainBook focuses on helping users through a single guided interaction. In the future, we would want it to remember past sessions, track recurring areas of difficulty, and help students reflect on how their reasoning improves over time. We would also continue testing the interface with more users so we could refine longer-term workflows and make the product feel even more natural for repeated use.